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1. Introduction

Organ Transplant: The Kidney Exchange Problem

Transplant Network: Despite the continuing and dramatic advances in medicine and health technology, the demand for organs for transplantation drastically exceeds the number of donors. To help this situation, US Congress passed the National Organ Transplant Act in 1984, establishing the Organ Procurement and Transplantation Network (OPTN) to match organ donors to patients with organ needs. Even with all this organizational technology and service in place, there are nearly 94,000 transplant candidates in the US waiting for an organ transplant and this number is predicted to exceed 100,000 very soon. The average wait time exceeds three years-double that in some areas, such as large cities. Organs for transplant are obtained either from a cadaver queue or from living donors. The keys for the effective use of the cadaver queue are cooperation and good communication throughout the network. The good news is that the system is functioning and more and more donors (alive and deceased) are identified and used each year with record numbers of transplants taking place every month. The bad news is that the candidate list grows longer and longer. Some people think that the current system with both regional and national aspects is headed for collapse with consequential failures for some of the neediest patients. Moreover, fundamental questions remain: Can this network be improved and how do we improve the effectiveness of a complex network like OPTN? Different countries have different processes and policies, which of these work best? What is the future status of the current system?

2. Assumptions

- There doesn't appear two or two more donators at the same moment;
- Not consider the sex difference between the donators and transplant candidates;
- The data found from web is precise enough;
- When the age of patient is over 65, don't consider his application ;
- Taking into account the kidney leave living body can live 48 hours ,so adapt the fastest conveyance to make sure the activity of kidney. And every transport can arrive at destination on time;
- Not consider the second kidney exchange;

3. Modeling and Results

3.1 Potential bottlenecks

3.1.1 Symbols

i : the order of the recipient in USA

j : region number

C_{ij} : the recipient whose region number is j and nation number is i

C_{ij}^w : the weight of C_{ij}

D_j : the donator whose region number is j

cD_j : whether appear the kidney which can be used to transplant (when there is no donator, $cD_j = 0$, else $cD_j = 1$)

N : current demand for kidney in USA

M : the number of the distributed region of USA

m : the number of determiner among characteristics and kidney transplant

n : the number of non-determiner among characteristics and kidney transplant

Z_{ij} : represent the matching level between C_{ij} and D_j

$R^{ij} = [r_{pq}^{ij}]_{p \times q}$: represent the matching matrix between C_{ij} and D_j

r_{pq}^{ij} : represent the relativity of every factor (refer to 3.1.3.1.i)

Y : the maximum in C_{ij}^w

3.1.2 Modeling

3.1.2.1 Model background

We consider kidney transplant problem: more and more recipient waiting for a limited number of the kidney supplied. According to a lot of data, we draw the quantity curve of donor and recipient (as Figure 1), these two curves appear potential bottlenecks, this means the rhythm of the number of candidate and the number of kidney is not consistent, and the bottlenecks problem will be more and more serious. At the same time, we found there is a used rate of the kidney supplied by contributor, it is an important standard, and it is the only one variable data we can control. In order to solve bottlenecks problem, we need to change the used rate of the kidney contributing to let bottlenecks grow narrower.

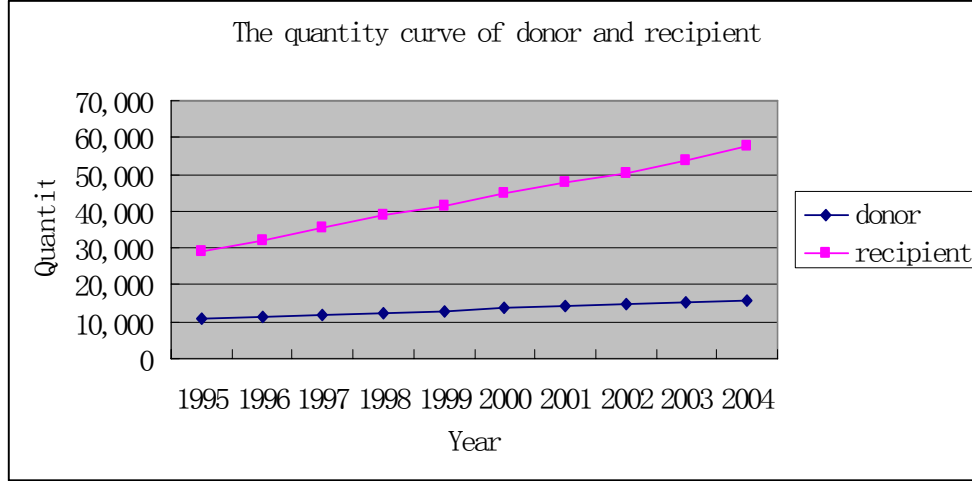


Figure 1. The quantity curve of donor and recipient

3.1.2.2 Weight Based Model

Corresponding to the policy of the Organ Procurement and Transplantation Network (OPTN), we establish an optimality model. We formulate the problem as the recipient maximize weight model:

$$Y = \text{Max} \left\{ C_{ij}^w \mid 0 \leq i < N, 0 \leq j < M, m \times n \leq Z_{ij} < m \times (n+1), Z_{ij} = \sum_{p=0}^{m+n-1} r_{pp}^{ij} \right\}$$

Y is the maximum of all weight of the recipient.

C_{ij}^w is the weight of the i th recipient whose region number is j . The bigger the weight, the faster the recipient obtains the kidney. So the weight is a important factor to be considered.

m is the number of determiner among characteristics and kidney transplant.

n is the number of non-determiner among characteristics and kidney transplant.

Z_{ij} is matching level between C_{ij} and D_j , the more illustrate the matching level is very better and the little illustrate the matching level is very worse. So this data is the more the better.

r_{pq}^{ij} is decided by expanded criteria donor and standard donor. And the indexes are mainly including blood type, PRA, HLA, Age and so on. If the indexes of the donator can match the indexes of transplant candidates are the same, define $r_{pq}^{ij} \neq 0$,

when $p = q$, and define $r_{pq}^{ij} = 0$, when $p \neq q$.

We consider the following two cases.

Case 1. The religion based model.

In this model, the U.S. was divided into 11 regions.

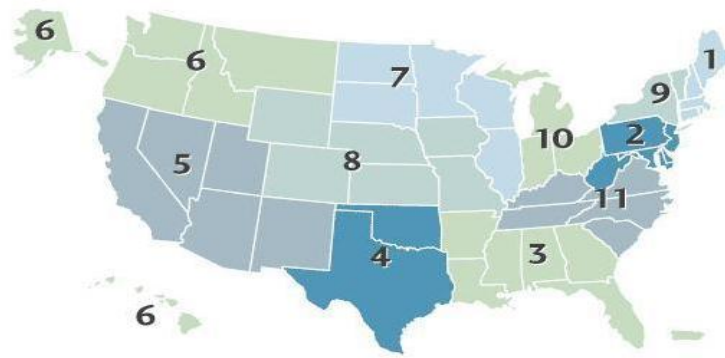


Figure 2. Region graph of U.S.

(See Policies 3.5.6.4) Members belongs to the Region in which they are located. The Regions are as follows:

Region 1 -Connecticut, Maine, Massachusetts, New Hampshire, Rhode Island, Vermont

Region 2 - Delaware, District of Columbia, Maryland, New Jersey, Pennsylvania, Northern Virginia, West Virginia

Region 3 - Alabama, Arkansas, Florida, Georgia, Louisiana, Mississippi, Puerto Rico

Region 4 - Oklahoma, Texas

Region 5 - Arizona, California, Nevada, New Mexico, Utah

Region 6 -Alaska, Hawaii, Idaho, Montana, Oregon, Washington

Region 7 -Illinois, Minnesota, North Dakota, South Dakota, Wisconsin

Region 8 -Colorado, Iowa, Kansas, Missouri, Nebraska, Wyoming

Region 9 - New York

Region 10 -Indiana, Michigan, Ohio

Region 11 -Kentucky, North Carolina, South Carolina, Tennessee, Virginia

Case 2.The state based model.

In this model, the U.S. was divided into 50 states.

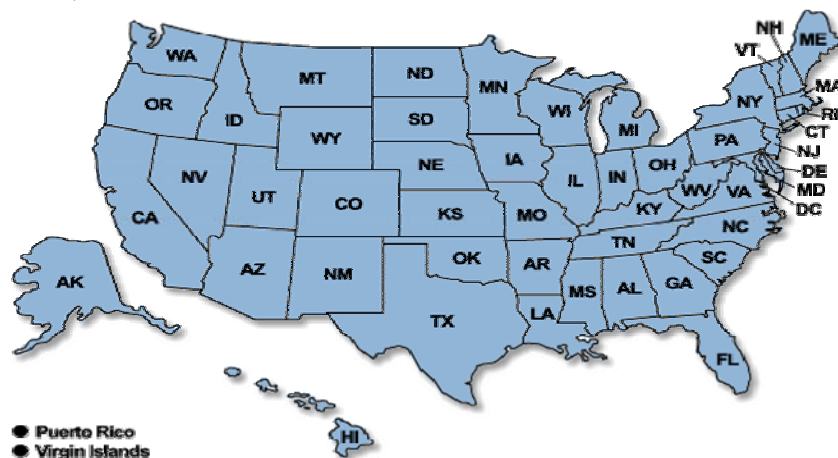


Figure 3. States Graph of U.S.

3.1.3 Results

3.1.3.1 Illustration with R^{ij} (a mutual compatibility matrix)

i. About mutual compatibility matrix

For each matching $\mu \in M$ and patient $i \in N$, $\mu(i) = i$ means that the patient i remains unmatched. For any matching $\mu \in M$ and pair of patients $i, j \in N$, $\mu(i) = j$ means that patient i receives a compatible kidney from a donor of patient j and patient j receives a compatible kidney from a donor of patient i . Since exchange is possible only among mutually compatible pairs, it is sufficient for our purposes to keep track of the symmetric $|N|$ by $|N|$ *mutual compatibility*

matrix $R = [r_{i,j}]_{i \in N, j \in N}$ defined by

$$r_{i,j} = \begin{cases} 1 & \text{if } j \succ_i i \text{ and } i \succ_j j, \\ 0 & \text{otherwise} \end{cases}$$

for any pair of (not necessarily distinct) patients $i, j \in N$, where $j \succ_i i$ means for any patient j with a compatible donor for patient i and $i \succ_j j$ means for any patient j without any compatible donor for patient i .

We will refer to the pair (N, R) as the *reduced problem* of $(N, \succ)$, where $\succ = (\succ_i)_{i \in N}$ denotes the list of patient preferences. Throughout the paper with the exception of the proofs of the results on incentives, we fix a problem $(N, \succ)$. Occasionally it will be helpful to think of the reduced problem as a graph $G = (N, R)$ whose vertices N are the patients (and their incompatible donors), and whose edges R are the connections between mutually compatible pairs of patients; i.e. there is an edge $i, j \in R$ if and only if $r_{i,j} = 1$. (It will be clear from the context whether R is the *mutual compatibility matrix* or the set of edges indicating mutual compatibility.) A matching then can be thought of as a subset of the set of edges such that each patient can appear in at most one of the edges. With this alternative representation if (i, j) is an edge in the matching μ , patients i and j are matched by μ and, if patient i does not appear in any

edge in the matching μ , she remains unmatched.

A *mechanism* is a systematic procedure that selects a matching for each problem.

ii. The process of handling R^{ij}

Matrix R^{ij} is a mutual compatibility matrix. On the basis of matching whether or not, evaluate r_{pq}^{ij} , when $p \leq m-1$, if can match (namely they are the same completely), so evaluate $r_{pq}^{ij} = n$, or $r_{pq}^{ij} = 0$; when $p > m$, if can not match (namely they are the same completely), evaluate $r_{pq}^{ij} = 1$, or $r_{pq}^{ij} = 0$. As to the different characteristic attribute evaluate $r_{pq}^{ij} = 0$.

3.1.3.2 Illustration with Z_{ij}

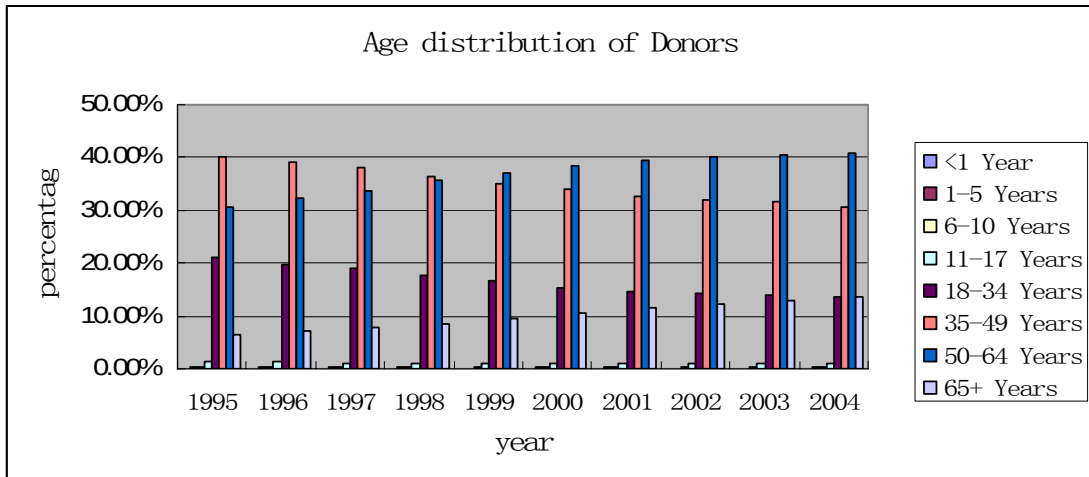


Table 1: Age distribution of Donors

Through appendix 2 data analysis, showing the distribution of different, as shown in Table 1. So age will be divided into four stages: $age \leq 18$; $18 < age \leq 34$; $34 < age \leq 50$; $age > 50$

Patient in different ages will be evaluated by one's vitality, operation success rate and contribution to society degree:

We according to the people's social value work out *Evaluation Standard*.

The process of handling Z_{ij} is as follow:

According to the formula $Z_{ij} = \sum_{p=0}^{m+n-1} r_{pp}^{ij}$, calculate every data of Z_{ij} .

Evaluation Standard:

$$0 < age \leq 18, weight = 0.2$$

$$18 \leq age < 34, weight = 0.4$$

$$34 \leq age < 50, weight = 0.3$$

$$age \geq 50, weight = 0.1$$

$$\text{relation} \begin{cases} \text{relatives} \begin{cases} age < 18 \\ 18 \leq age < 50 \\ age \geq 50 \end{cases} \\ \text{non-relatives} \begin{cases} age < 18 \\ 18 \leq age < 50 \\ age \geq 50 \end{cases} \end{cases}$$

3.1.3.3 IS Algorithm

We propose a heuristic algorithm, named Instantaneous Solve Algorithm, or call it IS Algorithm for short.

The steps of IS algorithm are as follows:

Step 1: Initialization: $D_j = 0$, $j = 1, 2, \dots, M$, according to the standard(age

standard ,3.1.3.2 give it), evaluate a *weight* with an initial data;

Step 2: When $cD_j = 1$, evaluate $j = k$, so $C_{ik}^w = C_{ik}^w + 1$;;

Step 3: For D_k , evaluate $flag = 1$, among all the demander. Search for all relatives in

the national scope, then evaluate $flag = 1$, $C_{ij}^w = C_{ij}^w + 2$. If not, then to Step 4 ;

Step 4: Calculate all R^{ij} , then get Z_{ij} ;

Step 5: If $Z_{ij} \geq m \times n$, carry out step 6. Or delete C_{ij} ;

Step 6: Judge exist present C_{ij} , if exist, put out C_{ij} which the weight is biggest(he is transplant candidates s) , $N = N - 1$. If not, give up transplanting.

Step7: When $flag = 1$, then to $C_{ij}^w = C_{ij}^w - 2$, $C_{ik}^w = C_{ik}^w - 1$

We draw a flow graph of the progress above, as Figure 4:

Flow Graph of IS Algorithm:

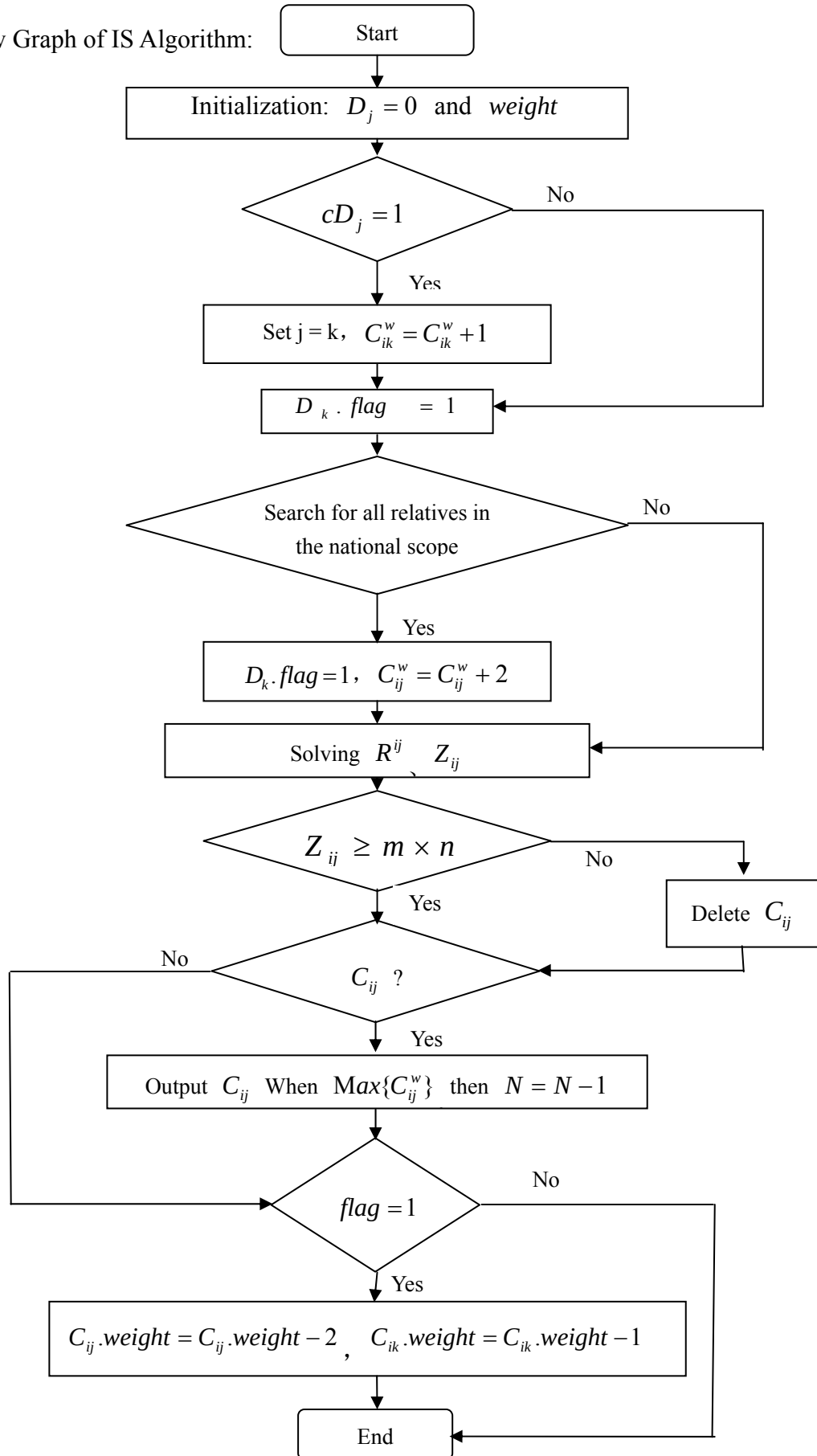


Figure 4. Flow Graph of IS Algorithm

In case1, M=11, put out the result by IS Algorithm Discussion.

In case2, M=50, put out the result by IS Algorithm Discussion.

3.1.3.4 Comparison between the two cases

Discuss these two models which is much better, that to say whether M is the bigger, the better?

Background (reference Policy 3.5.6):

Regional allocation: If a standard donor kidney is not accepted by any of the local transplant centers for local candidates, the kidney is to be allocated next via the regional list consisting of all candidates listed on the Waiting Lists of other Members within the same Region as the Member which procured the kidney. When a standard donor kidney is allocated regionally, it is to be offered to Members for specific candidates in the region according to the point system described in Policy 3.5.11 in descending point order beginning with the candidate in the region who has been assigned the highest number of points. With all regionally-shared standard donor kidneys, the Organ Center will advise the OPO for the transplant center for the candidate who has the highest number of points to seek alternate candidates within the OPO or other applicable Local Unit to receive the kidney in the event that the kidney cannot be used by the candidate. Selection of alternate candidates must be according to the point system for standard kidney allocation. If a local potential recipient(s) who has agreed to receive expanded criteria donor kidneys is not identified (*i.e.*, a match run and process for notifying the appropriate transplant program(s) initiated) within six hours post cross clamping of the donor aorta, the kidney is to be allocated next via the regional list consisting of all candidates who have agreed to receive expanded criteria donor kidneys listed on the Waiting Lists of other Members within the same Region as the Member which procured the kidney. When an expanded criteria donor kidney is allocated regionally, it is to be offered to Members for specific candidates in the region according to the point system described in Policy 3.5.12 in descending point order beginning with the candidate who has agreed to receive expanded criteria donor kidneys in the region who has been assigned the highest number of points.

National allocation.: If a standard donor kidney is not accepted by any transplant center in the Region in which the Member which procured the kidney is located, the kidney is to be allocated to Members for specific candidates in the other Regions nationally according to the point system described in Policy 3.5.11 in descending point order beginning with the candidate who has the highest number of points. With all nationally shared standard donor kidneys, the Organ Center will advise the OPO for the transplant center for the candidate who has the highest number of points to seek alternate candidates within the OPO or other applicable Local Unit to receive the kidney in the event that the kidney cannot be used by that candidate. Selection of alternate candidates must be according to the point system for standard donor kidney allocation. If an expanded criteria donor kidney is not accepted by any transplant center in the Region in which the Member which procured the kidney is located, the kidney is to be allocated to Members for specific candidates who have agreed to

receive expanded criteria donor kidneys in the other Regions nationally according to the point system described in Policy 3.5. 12 in descending point order beginning with the candidate who has the highest number of points.

Solve method:

We use diagram and some simple describe to explain, look at Figure 5.

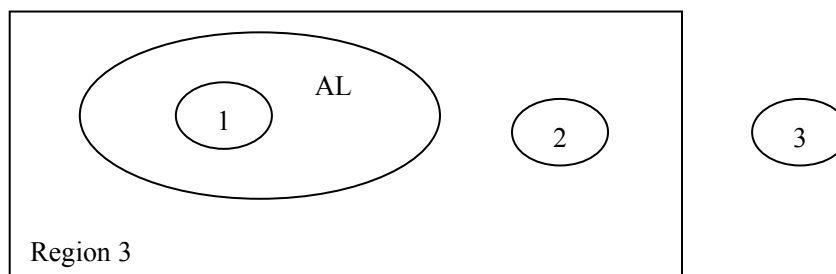


Figure 5. Part map of U.S (Region 3 and AL)

The local protection policy was pointed out in Policy 3.5.6, it says: all kidneys will be allocated first to local candidates as defined in Policy 3.1.7 the locale where the kidneys are procured. So we consider local protection policy is exist, and evaluate a weight to the same unit's area.

When adopt region model to allocate kidney, we consider 1 and 2 at first, evaluate a local protection weight each, then compare with 3. In this way, regardless the demanding urgency of 3 is strong; he will have no chance to get this kidney. But when adopt state model to allocate kidney, only 1 has local protection weight and compare with the region model, it appears more equitable.

In conclusion, M is the bigger, the model is the better.

3.2 Modified model based on other countries' policies

3.2.1 Symbols

i : the order of the recipient in USA

j : region number

C_{ij} : the recipient whose region number is j and nation number is i

C_{ij}^w : the weight of C_{ij}

D_j : the donator whose region number is j

cD_j : whether appear the kidney which can be used to transplant (when there is no donator, $cD_j = 0$, else $cD_j = 1$)

N : current demand for kidney in USA

M : the number of the distributed region of USA

m : the number of determiner among characteristics and kidney transplant

n : the number of non-determiner among characteristics and kidney transplant

Z_{ij} : represent the matching level between C_{ij} and D_j

$R^{ij} = [r_{pq}^{ij}]_{p \times q}$: represent the matching matrix between C_{ij} and D_j

r_{pq}^{ij} : represent the relativity of every factor (refer to 3.1.3.1.i)

Y : the maximum of match degree

3.2.2 Modeling

3.2.2.1 Model background

Investigate the transplantation policies used in a country other than the US. By modifying your model from Task 1, determine if the US policy be would improved by implementing the procedures used in the other country. As members of an expert analysis team (knowledge of public health issues and network science) hired by Congress to perform a study of these questions, write a one-page report to Congress addressing the questions and issues of Task 1 and the information and possible improvements you have discovered from your research of the different country's policies. Be sure to reference how you used your models from Task 1 to help address the issues.

Focusing on Kidney Exchange: Kidneys filter blood, remove waste, make hormones, and produce urine. Kidney failure can be caused by many different diseases and conditions. People with end-stage kidney disease face death, dialysis (at over \$60,000/yr), or the hope for a kidney transplant. A transplant can come from the cadavers of an individual who agreed to donate organs after death or from a live donor. In the US, about 68,000 patients are waiting for a kidney from a deceased donor, while each year only 10,000 are transplanted from cadavers and 6,000 from living individuals (usually relatives of the patients). Hence the median wait for a matching kidney is three years—unfortunately, some needy patients do not survive long enough to receive a kidney.

There are many issues involved in kidney transplantation—the overall physical and mental health of the recipient, the financial situation of the recipient (insurance for transplant and post-operation medication), and donor availability (is there a living donor willing to provide a kidney). The transplanted kidney must be of a compatible ABO blood type. The 5-year survival of the transplant is enhanced by minimizing the number of mismatches on six HLA markers in the blood. At least 2,000 would-be-donor/recipient pairs are thwarted each year because of blood-type incompatibility or poor HLA match. Other sources indicate that over 6,000 people on the current waiting list have a willing but incompatible donor. This is a significant loss to the donor population and worthy of consideration when making new policies and procedures. An idea that originated in Korea is that of a kidney exchange system,

which can take place either with a living donor or with the cadaver queue. One exchange is paired-kidney donation, where each of two patients has a willing donor who is incompatible, but each donor is compatible with the other patient; each donor donates to the other patient, usually in the same hospital on the same day. Another idea is list paired donation, in which a willing donor, on behalf of a particular patient, donates to another person waiting for a cadaver kidney; in return, the patient of the donor-patient pair receives higher priority for a compatible kidney from the cadaver queue. Yet a third idea is to expand the paired-kidney donation to 3-way, 4-way, or a circle (n-paired) in which each donor gives to the next patient around the circle. On November 20, 2006, 12 surgeons performed the first-ever 5-way kidney swap at Johns Hopkins Medical Facility. None of the intended donor-recipient transplants were possible because of incompatibilities between the donor and the originally intended recipient. At any given time, there are many patient-donor pairs (perhaps as many as 6,000) with varying blood types and HLA markers. Meanwhile, the cadaver queue receives kidneys daily and is emptied daily as the assignments are made and the transplants performed.

3.2.2.2 Matching Model

According to the policy of the Organ Procurement and Transplantation Network (OPTN), we establish an optimality model. We formulate the problem as the matching level of the donator and recipient model 2:

$$Y = \text{Max} \left\{ Z_{ij} \mid 0 \leq i < N, 0 \leq j < M, m \times n \leq Z_{ij} < m \times (n+1), Z_{ij} = \sum_{p=0}^{m+n-1} r_{pp}^{ij} \right\}$$

Y is the maximum of all match level of the donator and recipient.

i is the order of the recipient in USA.

j is region number.

m is the number of determiner among characteristics and kidney transplant

n is the number of non-determiner among characteristics and kidney transplant

Z_{ij} is matching level between C_{ij} and D_j , the more illustrate the matching level is

very better and the little illustrate the matching level is very worse. So this data the more the better.

r_{pq}^{ij} is decided by expanded criteria donor and standard donor. And the indexes are mainly including blood type, PRA, HLA, Age and so on. If the indexes of the donator can match the indexes of transplant candidates are the same, define $r^{ij} \neq 0$, when $p = q$, and define $r^{ij} = 0$, when $p \neq q$

3.2.3 Results

3.2.3.1 Illustration with R^{ij}

The process of handling R^{ij} is as follow:

Matrix R^{ij} is a *mutual compatibility matrix*. On the basis of matching whether or not, evaluate r_{pq}^{ij} , when $p \leq m-1$, if can match (namely they are the same completely), so evaluate $r_{pq}^{ij} = n$, or $r_{pq}^{ij} = 0$; when $p > m$, if can not match (namely they are the same completely), evaluate $r_{pq}^{ij} = 1$, or $r_{pq}^{ij} = 0$. As to the different characteristic attribute evaluate $r_{pq}^{ij} = 0$.

3.2.3.2 Illustration with Z_{ij}

The process of handling Z_{ij} is as follow:

According to the formula $Z_{ij} = \sum_{p=0}^{m+n-1} r_{pp}^{ij}$, calculate every data of Z_{ij} .

Evaluation Standard:

$0 < age \leq 18, weight = 0.2$

$18 \leq age < 34, weight = 0.4$

$34 \leq age < 50, weight = 0.3$

$age \geq 50, weight = 0.1$

$$\text{relation} \begin{cases} \text{relatives} \begin{cases} age < 18 \\ 18 \leq age < 50 \\ age \geq 50 \end{cases} \\ \text{non-relatives} \begin{cases} age < 18 \\ 18 \leq age < 50 \\ age \geq 50 \end{cases} \end{cases}$$

3.2.3.3 IS- II Algorithm

In order to solve this problem (model 2), we expand IS Algorithm, get Instantaneous Solve II Algorithm, or call it IS- II Algorithm for short.

The steps of IS II Algorithm are as follows:

Step 1: Initialization: $D_j = 0$, $j = 1, 2, \dots, M$, according to the standard (age

standard ,3.1.3.2 give it), evaluate a *weight* with an initial data;

Step 2: When $cD_j = 1$, value $j = k$, so $C_{ik}^w = C_{ik}^w + 1$;

Step 3: As to D_k , evaluate $flag = 1$, among all the demander. Search for all relatives in the national scope, then evaluate $flag = 1$, $C_{ij}^w = C_{ij}^w + 2$. If not, then to Step 4 ;

Step 4: Calculate all R^{ij} , then get Z_{ij} ;

Step 5: If $Z_{ij} \geq m \times n$, carry out step 6. Or delete C_{ij} ;

Step 6: Judge exist present C_{ij} , if exist, put out C_{ij} which the data of Z_{ij} is biggest (he is transplant candidates s), $N = N - 1$. If not, give up transplanting.

Step 7: When $flag = 1$, then to $C_{ij}^w = C_{ij}^w - 2$, $C_{ik}^w = C_{ik}^w - 1$

Step 8: End.

3.2.4 Report

According to national conditions of USA, we know that all states in the United States is relatively independent in the political, legal, cultural and any other areas. in the issue of Kidney transplantation, the United States policy consider every region is a unit (each state can be seen as a unit). Carry out the area protecting system. It is unreasonable for the kidney transplant, because it is not better to achieve the principles of fairness and impartiality, at the same time affects the efficiency of kidney transplantation. For other countries, implement policies about kidney transplantation is equal to every region, and this policy will be better to achieve the principles of fairness and impartiality, at the same time improve the efficiency of a kidney transplantation. This means even if the United States carried out area protecting system, to limit the area of geographical regions. At the same time, the kidney not been matched in this region will be used in other region. According to the organ transplant policies for the United States, We believe the policy choices: kidney recipients are based on the importance of the receiver. In order to solve this problem we have adopted policies of other countries, through the matching the donor and recipients to choose which is best.

In American, carrying out a uniform national standard is a bit difficult, because of national conditions of USA. Since all states are relatively independent, resources kidney can not be used in best. And it is difficult to solve the bottleneck problem. In the States, it's hard to carry out National allocation, because American is a country with more than the population of the area, without adopt area protecting system, will

the sum of the cost. Of course, in the model, we adopt Instantaneous Solution: when appear one donor match and distribute it at once. This program will be able to ease the bottleneck in kidney transplantation

3.3 Effect of exchanges

3.3.1 Symbols

i : the order of the donator in USA.

j : region number (it aimed to i)

k : the order of the recipient in USA

t : region number (it aimed to k)

e_{ik} : the edge which from i to k

a_{ik} : the weight of edge in region

b_{ij} : the weight of edge in USA

x_{ik} : judge whether edge is exist, if exist value it 0, or value it 1

C_{kt} : the recipient whose region number is k and the nation number is t

C_{kt}^w : the weight of C_{kt}

D_{ij} : the donator whose region number is j and the nation number is i

N : current demand for kidney in USA

P : current donor for kidney in USA

M : the number of the distributed region of USA,

m : the number of determiner among characteristics and kidney transplant

n : the number of non-determiner among characteristics and kidney transplant

Z_{ik} : represent the matching level between C_{kt} and D_{ij}

$R^{ik} = [r_{pq}^{ik}]_{(m+n) \times (m+n)}$: represent the matching matrix between C_{kt} and D_{ij}

r_{pq}^{ik} : represent the relativity of every factor

W_j : the weight of recipient

Quality : the sum of the demand weight of people who required

Number : the number of kidney donate which can be matched successfully

Benefit : the benefit of the successful operation (taking into account the medical and

psychological dynamics of the situation)

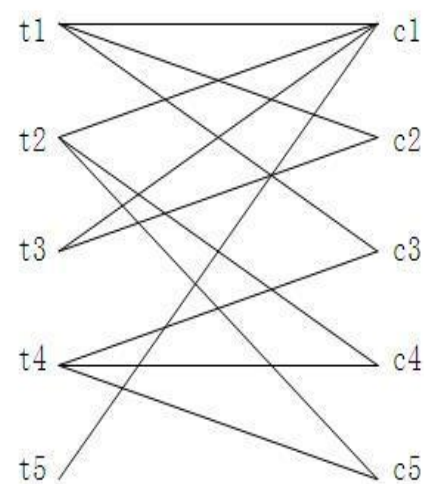
3.3.2 Modeling

3.3.2.1 Model background

Devise a procedure to maximize the number and quality of exchanges, taking into account the medical and psychological dynamics of the situation. Justify in what way your procedure achieves a maximum. Estimate how many more annual transplants your procedure will generate, and the resulting effect on the waiting list.

About bipartite graphs: The present paper explores how to organize such exchanges. The theoretical groundwork for kidney exchange without a constraint on the size of exchanges, explored in Roth et al, was laid in the papers by Shapley and Scarf, Roth and Postlewaite, Roth, and Abdulkadiro lu and SÖnmez. The constrained exchange problem studied in the present paper is closely related to elegant results from graph theory, which will prove very useful, in ways recently pioneered by Bogomolnaia and Moulin. They looked at a problem of pair wise matching with 0–1 preferences on two-sided graphs, i.e. on graphs in which the parties to the exchange can be a priori divided into two sets, e.g. into buyers and sellers, each of which can only trade with the other. Bipartite graphs cannot be used to model kidney exchange, since any patient–donor pair might potentially exchange with any other. The present paper therefore generalizes such a model to the case of arbitrary graphs.

Our paper builds on the closely related recent paper by Bogomolnaia and Moulin. They considered two-sided matching, i.e. matching between two sets of agents that can be specified exogenously (e.g. firms and workers), such that an agent on one side of the market can only be matched with an agent on the other side, modeled as a bipartite graph, with 0–1 preferences. It was their paper that made us aware of some of the graph-theoretic results that we also use here. Our results on the egalitarian mechanism generalize their corresponding results to general, not necessarily bipartite graphs. Kidney exchange cannot be modeled as a two-sided market, since any patient with incompatible donors can potentially be matched with any other. 11 The extensions to the general case are of interest not only because of the importance of the application to kidney exchange, but also because of the insights they give into pair wise exchange in general graphs, and the technical challenges that the generalization to arbitrary graphs presents.



Example of Bipartite Graph

Assignment problem: Assignment problem is generally recognized to be the beginning of the development of practical solution methods for and variations on the classic assignment problem was the publication on the Hungarian method for its solution. Recognizing that there are researchers who have an interest in some variation on the classic AP, but do not know what the standard name of that variation is, thus making it more difficult to search for previous work on the topic, this paper is intended to provide a reasonably comprehensive survey of the different variations on the assignment problem. The purpose here is not to identify the best solution procedure for each of these “adjective” APs. Assignment problem have several classic models. In this paper used the classic assignment problem recognizing agent qualification.

The classic assignment problem recognizing agent qualification:

In their work on a particular version of the assignment problem with side constraints, Caron et al use a mathematical model for a variation of the classic AP in which there are m agents and n tasks, not every agent is qualified to do every task, and the objective is utility maximization:

$$\begin{aligned} &\text{Maximize} \quad \sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij} \\ &\text{Subject to:} \quad \sum_{i=1}^m q_{ij} x_{ij} \quad j = 1, 2, \dots, n \\ &\quad \quad \quad \sum_{j=1}^n q_{ij} x_{ij} \quad i = 1, 2, \dots, m \\ &\quad \quad \quad x_{ij} = 0 \quad \text{or} \quad 1 \end{aligned}$$

where $x_{ij} = 1$ if agent i is assigned to task j , 0 if not, $q_{ij} = 1$ if agent i is qualified to perform task j , 0 if not, and c_{ij} = the utility of assigning agent i to task j (with $c_{ij} = 0$ if $q_{ij} = 0$). The first set of constraints ensures that no more than one qualified agent is assigned to any task and the second set of constraints ensures that no agent is assigned to more than one task. Note that even if m is greater than or equal to n it may not be possible to assign a qualified agent to every task or to give every agent a task for which it is qualified. Minimization problems can be handled using either of the methods suggested above for handling maximization problems for the standard classic AP.

Strategies: Patients can face agonizing choices. For example, suppose a barely compatible-in terms of HLA mismatches-kidney becomes available from the cadaver queue. Should they take it or wait for a better match from the cadaver queue or from an exchange? In particular, a cadaver kidney has a shorter half-life than a live donor kidney.

3.3.2.2 The multi objective programming

This is a multiple objectives assignment problem. We establish an optimality model. We formulate the problem as the *multi objective programming* model:

$$\begin{cases} \text{Max} & \text{Number} = \sum_{i=0}^{P-1} \sum_{k=0}^{N-1} x_{ik} \\ \text{Max} & \text{Quality} = \sum_{i=0}^{P-1} \sum_{k=0}^{N-1} x_{ik} W_k \\ \text{Max} & \text{Benefit} = \sum_{i=0}^{P-1} \sum_{k=0}^{N-1} x_{ik} Z_{ik} \end{cases}$$

$$\text{s.t.} \quad \begin{cases} m \times n \leq Z_{ik} \leq m \times (n+1) \\ 0 \leq \sum_{k=0}^{N-1} x_{ik} \leq 1 \quad i = 0, 1, \dots, P-1 \\ 0 \leq \sum_{i=0}^{P-1} x_{ik} \leq 1 \quad k = 0, 1, \dots, N-1 \\ x_{ik} = 0 \quad \text{or} \quad 1 \end{cases}$$

Quality is the sum of the demand weight of people who required .

Number is the number of kidney donate which can be matched successfully.

Benefit is the benefit of the successful operation (taking into account the medical and psychological dynamics of the situation).

The *multi objective programming* model's solution procedure describe as follow:

1. We make the max of the *Number* .
2. To assurance the max of the *Number* ,we make the max of the *Benefit* .
3. To assurance the max of the *Number* and *Benefit* ,we make the max of *Quality* .

3.3.3 Results

3.3.3.1 Illustration with R^{ij}

r_{pq}^{ik} is decided by expanded criteria donor and standard donor. And the indexes are mainly including blood type, PRA, HLA, Age and so on. If the indexes of the donator can match the indexes of transplant candidates are the same, define $r_{pq}^{ik} \neq 0$, when $p = q$, and define $r_{pq}^{ik} = 0$, when $p \neq q$.

The process of handling R^{ik} is as follow:

Matrix R^{ik} is a mutual compatibility matrix. On the basis of matching whether or not, evaluate r_{pq}^{ik} , when $p \leq m-1$, if can match(namely they are the same

completely) , so evaluate $r_{pq}^{ik} = n$, or $r_{pq}^{ik} = 0$; when $p > m$, if can not match(namely they are the same completely) , evaluate $r_{pq}^{ik} = 1$, or $r_{pq}^{ik} = 0$. As to the different characteristic attribute evaluate $r_{pq}^{ik} = 0$.

3.3.3.2 Illustration with Z_{ij}

The process of handling Z_{ik} is as follow:

According to the formula $Z_{ik} = \sum_{p=0}^{m+n-1} r_{pp}^{ik}$, calculate every data of Z_{ik}

The process of handling Z is as follow:

The matrix $Z = [Z_{ik}]_{P \times N}$ is all represent the matching stage between D_{ij} and C_{kt} .

On the basis of matching stage, evaluate Z_{ik} .

3.3.3.3 Solutions about a_{ik}

The process of solving a_{ik} is as follow:

a_{ik} is D_{ij} at the j region's weight. When $j = t$ and i is constant, $count(Z_{ik} \geq m \times n)$,

$$a_{ik} = \frac{1}{count(Z_{ik} \geq m \times n)}$$

3.3.3.4 Illustration with b_{ij}

The process of handling b_{ij} is as follow:

b_{ij} is D_{ij} at USA's weight. When i is constant, $count(Z_{ik} \geq m \times n)$,

$$b_{ij} = \frac{1}{count(Z_{ik} \geq m \times n)}$$

Evaluation Standard:

$0 < age \leq 18, weight = 0.2$

$18 \leq age < 34, weight = 0.4$

$34 \leq age < 50, weight = 0.3$

$age \geq 50, weight = 0.1$

3.3.4 MIS Algorithm

In order to solve this model we design a new heuristic algorithm More Instantaneous Solve Algorithm, call it MIS Algorithm for short.

Explained MIS Algorithm as follow:

Step 1: Create the bipartite graphs, $D_{ij} \in V_1, C_{kt} \in V_2, e_{ik} \in E$.

Step 2: Create M bipartite graphs, when $j = t$, then $D_{ij} \in V_{j1}, C_{kt} \in V_{j2}, e_{ik} \in E_j$.

Step 3: according to the standard, evaluate C_{kt}^w with an initial data;

Step 4: Initialization: $z_{ik} = 0, e_{ik} = 0, x_{ik} = 0$

Step 5: Respectively for majorizing collocation of every bipartite graphs, the concrete steps is as follow:

Step 5.1: calculate represent the matching stage between D_{ij} and C_{kt} , get the

matrix $Z = [z_{ik}]_{P \times N}$.

Step 5.2: calculate represent a_{ik} and b_{ij} .

Step 5.3: when $j = t$, a_{ik} and C_{kt}^w by DESC, select the best benefit to match

(calling the IS II Algorithm), delete D_{ij} and C_{kt} , made $x_{ik} = 1$, if

there isn't best edge, then to Step 5.3, else if Step 6 have been executed, then to Step 7, else then to Step 6.

Step 6: Create new bipartite graphs $D_{ij} \in V_1, C_{kt} \in V_2, e_{ik} \in E$ then to Step 5.

Step 7: End.

Flow Graph of MIS Algorithm:

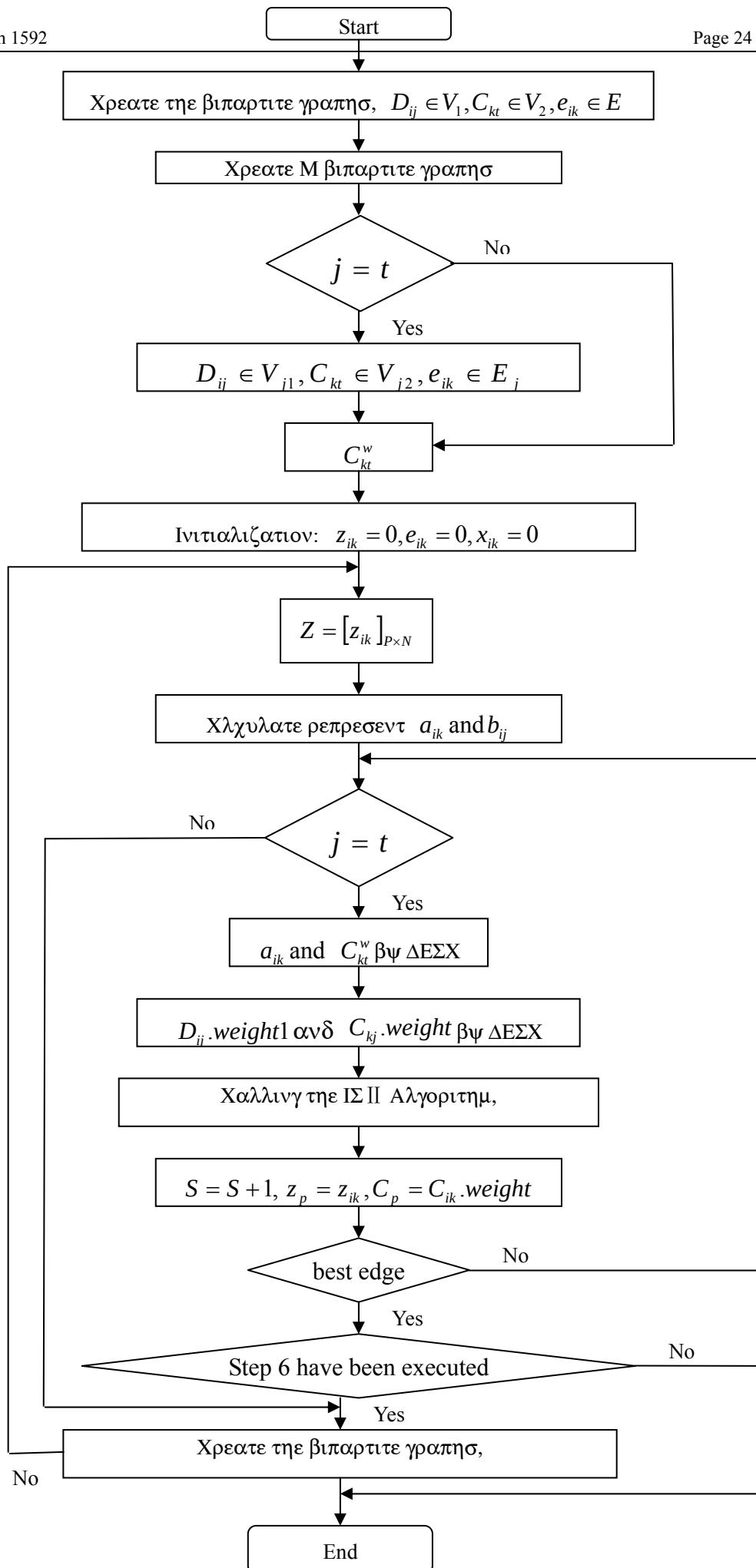


Figure 7. Flow Graph of MIS Algorithm

3.4 The recipient selection problem

3.4.1 Symbols

i : Sequence of donator over the whole country,

j : Sequence of recipient over the whole country,

D_i : The donator of sequence is j

D_i has three attributes: Character ,type and weight. And Character including age, ABO, HLA, PRA (stand for the information about relatives) and so on .Type including Living Donor (LD) and Deceased Donor (DD)

D_i^t :The type of the donator, including Living Donor (LD) and Deceased Donor (DD).

D_i^w :The weight of the donator.

C_j : The recipient of sequence is i .

C_j has one attribute: characteristic vector ($Character[m+n]$)

N : Current demand for kidney in USA,

M : Current donor for kidney in USA,

M : The number of determiner among characteristics and kidney transplant (the data must be equal),

n : The number of non-determiner among characteristics and kidney transplant (the data may be equal),

Z_{ij} : The matching stage between D_i and C_j ,

$R^{ij} = [r_{pq}^{ij}]_{(m+n) \times (m+n)}$: The matching matrix between D_i and C_j ,

r_{pq}^{ij} : The relativity of every factor.

S_{ij} : The selecting stage between D_i and C_j ,

S_{avg} : The average of the selecting stage in the USA.

S_j : The average of the selecting stage with C_j .

S_p : The select standard of the selecting stage

S : The matrix of the selecting stage

Z : The matrix of the matching stage

count1 : The all number of according with the matching.

count2 : The number of according with the matching with C_j .

P :the recipient not to select the kidney , the probability that recipient can choose kidney which match better ones latter.

3.4.2 Model background

Devise a strategy for a patient to decide whether to take an offered kidney, or to even participate in a kidney exchange. Consider the risks, alternatives, and probabilities in your analysis.

Ethical Concerns: Transplantation is a controversial issue with both technical and political issues that involve balancing what is best for society with what is best for the individual. Criteria have been developed very carefully to try to ensure that people on the waiting list are treated fairly, and several of the policies try to address the ethical concerns of who should go on to the list or who should come off. Criteria involved for getting on or coming off the list can include diagnosis of a malignant disease, HIV infection or AIDS, severe cardiovascular disease, a history of non-compliance with prior treatment, or poorly controlled psychosis. Criteria used in determining placement priority include: time on the waiting list, the quality of the match between donor and recipient, and the physical distance between the donor and the recipient. As a result of recent changes in policy, children under 18 years of age receive priority on the waiting list and often receive a transplant within weeks or months of being placed on the list. The United Network for Organ Sharing Website recently (Oct 27, 2006) showed the age of waiting patients as:

Under 18: 748
 18 to 34: 8,033
 35 to 49: 20,553
 50 to 64: 28,530
 65 and over: 10,628

One ethical issue of continual concern is the amount of emphasis and priority on age to increase overall living time saved through donations. From a statistical standpoint, since age appears to be the most important factor in predicting length of survival, some believe kidneys are being squandered on older recipients. Political issues: Regionalization of the transplant system has produced political ramifications (e.g., someone may desperately need a kidney and is quite high on the queue, but his or her deceased neighbor's kidney still can go to an alcoholic drug dealer 500 miles away in a big city). Doctors living in small communities, who want to do a good job in transplants, need continuing experience by doing a minimum number of transplants per year. However, the kidneys from these small communities frequently go to the hospitals in the big city and, therefore, the local doctors cannot maintain their proficiency. This raises the question, should transplants be performed only in a few large centers, by a few expert and experienced surgeons? Would that be a fair system and would it add or detract from system efficiency? Many other ethical and political

issues are being debated. Some of the current policies can be found at <http://www.unos.org/policiesandbylaws/policies.asp?resources=true>. For example, recent laws have been passed in the US that forbid the selling or mandating the donation of organs, yet there are many agencies advocating for donors to receive financial compensation for their organ. The state of Illinois has a new policy that assumes everyone desires to be an organ donor (presumed consent) and people must opt out if they do not. The Department of Health and Human Services Advisory Committee on Organ Transplantation is expected to recommend that all states adopt policies of presumed consent for organ donation. The final decision on new national policies rests with the Health Resources and Services Administration within the US Department of Health and Human Services.

3.4.3 Criterion model and TS Algorithm

This model is established to solve a judgment problem. Refer to the policy of the Organ Procurement and Transplantation Network (OPTN), we establish a Criterion model about the recipient select kidney's question.

$$P = \frac{\text{count2} - i}{\text{count2}}$$

i represent the recipient accept i th kidney.

Because of this judgment model, we create a heuristic algorithm, named Time Solve Algorithm, and call it TS Algorithm for short.

The concrete steps of TS algorithm is as follow:

Step 1: Initialization: $\text{count1} = \text{count2} = 0$;

Step 2: calculate represent the matching stage between D_i and C_j , used the

$$\text{matrix } Z = [Z_{ij}]_{M \times N}.$$

Step 3: if $Z_{ij} \geq m \times n$, then $\text{count1} = \text{count1} + 1, S_{ij} = Z_{ij} \times D_i^w, \text{SUM}(S_{ij})$; else $S_{ij} = 0$

Step 4: $S_{\text{avg}} = \text{AVG}(S_{ij})$

Step 5: Determined C_j (j is constant), if $S_{ij} \neq 0$, then

$$\text{count2} = \text{count2} + 1, \text{SUM}(S_{ij})$$

Step 6: $S_j = \text{AVG}(S_{ij})$, if $S_{\text{avg}} > S_j$, then $S_p = S_{\text{avg}}$, else $S_p = S_j$,

Step 7: According to the time of donator donor kidney by ASC, if $S_{ij} > S_p$, then

C_j accepted kidney, else then to Step 7;

Step 8: End

r_{pq}^{ij} is decided by expanded criteria donor and standard donor. And the indexes are mainly including blood type, PRA, HLA, Age and so on. If the indexes of the donor can match the indexes of transplant candidates are the same, define $r_{pq}^{ij} \neq 0$, when $p = q$, and define $r_{pq}^{ij} = 0$, when $p \neq q$.

The process of handling R^{ij} is as follow:

Matrix R^{ij} is a mutual compatibility matrix. On the basis of matching whether or not, evaluate r_{pq}^{ij} , when $p \leq m-1$, if can match (namely they are the same completely), so evaluate $r_{pq}^{ij} = n$, or $r_{pq}^{ij} = 0$; when $p > m$, if can not match (namely they are the same completely), evaluate $r_{pq}^{ij} = 1$, or $r_{pq}^{ij} = 0$. As to the different characteristic attribute evaluate $r_{pq}^{ij} = 0$.

The process of handling Z_{ij} is as follow:

According to the formula $Z_{ij} = \sum_{p=0}^{m+n-1} r_{pp}^{ij}$, calculate every data of Z_{ij} .

For ensure the data of D_i^w , we consider the type of the donor. From the data of Appendix 3, we introduce the survival of the Living donor and the deceased donor at 2003.

$$\begin{cases} D_i^t = LD, D_i^w = 0.953 \\ D_i^t = DD, D_i^w = 0.89 \end{cases}$$

Flow Graph of TS Algorithm:

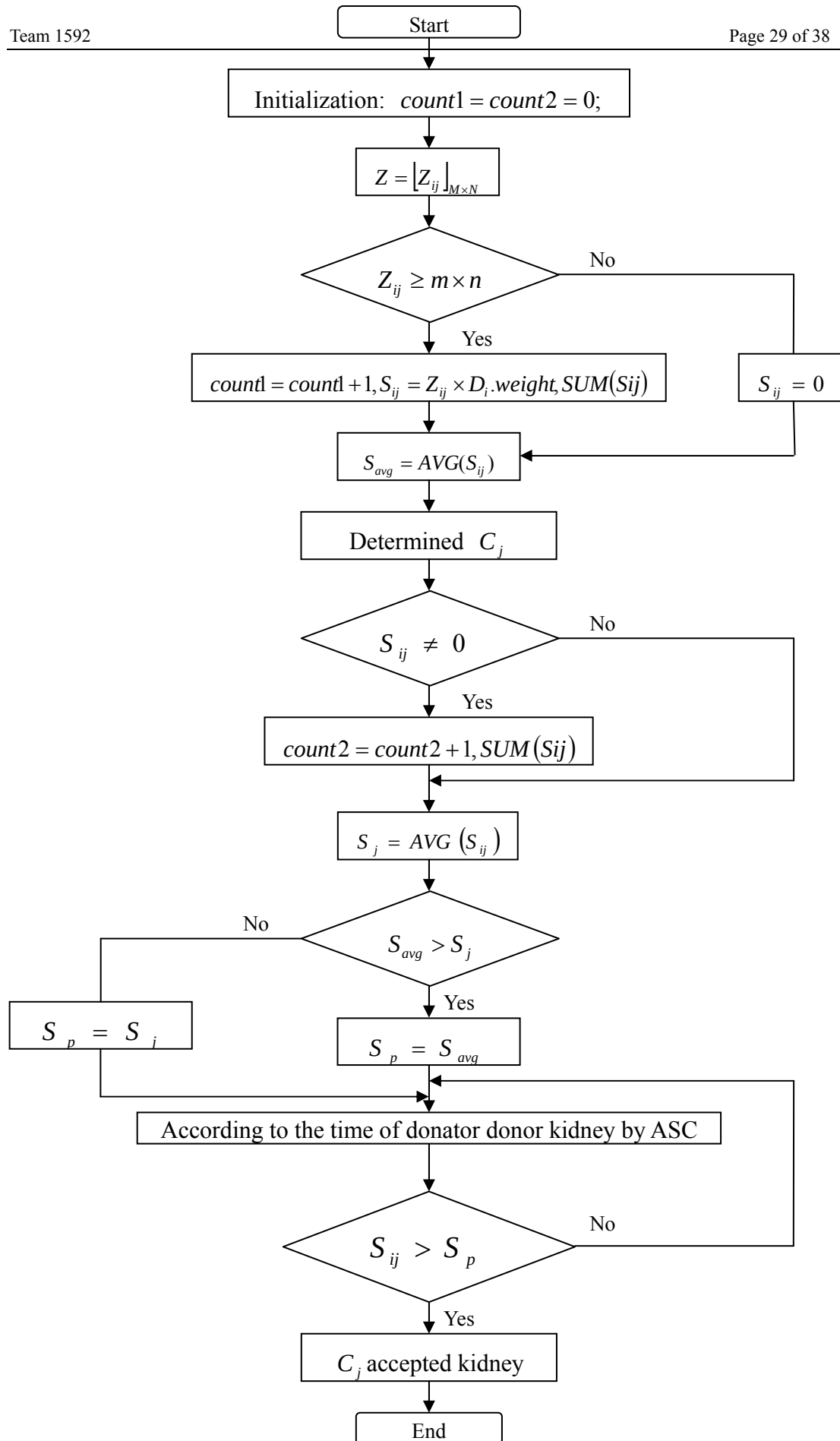


Figure 7. Flow Graph of TS Algorithm

3.5 Recommendation and suggestions

3.5.1 Problem background

Based on your analysis, do you recommend any changes to these criteria and policies? Discuss the ethical dimensions of your recommended exchange procedure and your recommended patient strategy (Tasks 3 and 4). Rank order the criteria you would use for priority and placement, as above, with rationale as to why you placed each where you did. Would you consider allowing people to sell organs for transplantation? Write a one-page paper to the Director of the US Health Resources and Services Administration with your recommendations.

3.5.2 Opinions and suggests

Without a doubt, organ transplantation is one of the wondrous medical developments of the century. Organ transplant surgery has given life where death was once a certainty for tens of thousands of chronically ill Americans. Every patient who survives transplant surgery to lead a productive life has been touched by a miracle.

But like all miracles, these are in short supply. Every day, eleven Americans whose lives could be saved by transplantation die because there are not enough organs for everyone who needs them. Increasing organ donation rates must be our number one priority.

Unfortunately, only 5,500 cadaveric donors, about one half to one third of all potential donors, contribute organs for transplantation. While the number of cadaveric organ donors has increased by 30 percent, the number of deaths on the waiting lists has also grown, by nearly 177 percent.

Allocating scarce organs to patients on the basis of geography is fundamentally unfair. Not only is current allocation policy inequitable, existing allocation boundaries simply do not make sense. The transplant network is using boundaries that were designed for organ procurement. These boundaries are not based on medical criteria and I believe it would be a mistake to continue using them for allocation purposes. Transplant centers have adapted to the current system, and in fact, have proliferated under it. But patients have not. Regardless of any economic gains for local transplant centers, the system is bad for patients.

Who should receive the organ?

Under the current policy, patients who are less ill receive transplants while more severely ill patients, perhaps only a few miles away, die. We all have friends and relatives who live in different parts of the country. Should any one of them have a better chance of living than the other if they needed an organ transplant? Is it fair that patients with virtually identical medical needs are treated differently solely because of where they live?

The unfairness exists, not only between different parts of the country, but even within states. These waiting list disparities are the most visible shortcoming of the current system. Less visible are the resulting inequities among those who receive

organs. Where waiting times are the shortest, organs may go to patients who are less ill; while at the same moment, in areas where patients wait longer, organs often are not offered to patients with greater medical need. In the worst cases, patients die in areas where waiting times are long, while, at the same time, organs are being made available to patients who are less ill in areas with shorter waiting times.

So in our opinion, we consider the rule contains three common sense goals that are consistent with the equity requirements of the National Organ Transplant Act. The first goal requires that criteria for placing patients on waiting lists be standardized and be based as much as possible on objective medical criteria. The network cannot be truly fair if the standards for placing patients on waiting lists differ across the country. UNOS agrees with this goal and is working toward it.

The second goal requires that criteria for determining the medical status of patients be standardized, again on the basis of objective criteria. Uniform criteria for determining medical status will prevent gaming of the system. UNOS also agrees with this goal and is working toward its accomplishment.

The third goal requires that medical urgency, not geography, be the main criterion - not the sole criterion - for allocating organs. This goal also meets the intent of the Organ Transplant Act for an equitable nationwide organ transplant network. UNOS objects to this goal, and if their position prevails, the current inequitable system will be maintained.

There is a central purpose to the performance goals, which is to ensure, to the maximum extent possible, that all patients, regardless of where they live, are treated the same. Because there are not yet enough organs to save everyone on the waiting lists, I believe we must, at a minimum, guarantee fairness, while we continue to strive to increase donation.

About selling organs:

Selling organs is again at issue.³ The idea is that people should be given motivation to sell organs at a fixed price set by an agency. The organs are then donated to recipients. There are many controversial issues around this. From whom would this company buy organs and at what price? If the company sought organs from a poorer nation, this may reduce the black market, but on the other hand it may appear that rich countries exploit poorer ones. This can be resolved by not allowing countries to buy outside their own territory. This would on the one hand not resolve the question of black marketing and on the other still exploit people in need or in debt. Would a rich man sell his kidney? And why should he get money for it if he is already rich? Why should he not donate instead? The whole idea of buying organs, even if they are to be bought by the government and donated free to recipients, creates considerable--for many, insurmountable--ethical objections on who sells them. It goes without saying that doctors should not participate in or encourage the buying or selling of organs until such issues are ironed out.

This is some of our opinions on basis of the models we made.

3.6 Discussions with respect to transplantation

3.6.1 Problem background

From the potential donor's perspective, the risks in volunteering involve assessing the probability of success for the recipient, the probability of survival of the donor, the probability of future health problems for the donor, the probability of future health risks (such as failure of the one remaining kidney), and the postoperative pain and recovery. How do these risks and others affect the decision of the donor? How do perceived risks and personal issues (phobias, irrational fears, misinformation, previous experiences with surgery, level of altruism, and level of trust) influence the decision to donate? If entering a list paired network rather than a direct transplant to the relative or friend, does the size n of the n -paired network have any effect on the decision of the potential donor? Can your models be modified to reflect and analyze any of these issues? Finally, suggest ways to develop and recruit more altruistic donors.

3.6.2 Discussions

Proposal of encourage a citizen to donate kidney

From information we obtained, it is found that every day in the United States, sixteen people on organ donation waiting lists die waiting for a way to prolong their lives.

There are two ways to be an organ donor. More commonly, organs come from people who have expressed a wish to donate their organs after their death. These donors (sometimes referred to as cadaveric or non-living donors) notify their family and friends of their decision and fill out an organ donor card so that after they die, their organs can prolong the life of others. People filling out an organ donor card can specify which organs they wish to donate. Hearts, livers, lungs, kidneys, pancreas, intestines, corneas, skin, connective tissue, and bone marrow all can be donated. But there is also a issue that though the system is functioning and more and more donors (alive and deceased) are identified and used each year with record numbers of transplants taking place every month, the candidate list still grows longer and longer. There is a without argued fact (the success rate of living kidney transplants is higher than one of cadaver transplants obviously) that successful of living kidney transplants will be. So adapt living kidney transplants is a best approach to restore patient.

In fact, because of phobias, irrational fears, misinformation, previous experiences with surgery, level of altruism, and level of trust, a lot of people afraid of donating kidney.

But the benefits for living donor recipients have been well documented. The relative low risk of physical complications during donation has also been well documented. Less well understood is the psychosocial risk to donors. Most published reports have indicated an improved sense of well-being and a boost in self-esteem for

living kidney donors. A lot of information can be seen in web net. At the same time, patients will also get benefit, It has been shown that the chances of a successful transplant are better with living donor organ than with cadaveric kidneys. Patients are spared the often grueling years of waiting spent on hem dialysis (blood cleansing) or continuous ambulatory peritoneal dialysis (CAPD).

Kidney dialysis normally reduces the patient's fitness for work by half, which means changes at the workplace and lower career prospects. The patient's diet must consist of are an extremely low-salt foods and minimal fluid intake. The need to choose a destination with dialysis facilities restricts the patient's opportunities to travel or go away on vacation. It is also common for the patient's relationship to suffer, since sexual problems may arise due to reduced libido or impotence.

Plus, living kidney transplants are more successful, and the closer the blood relationship, the better the results. Surprisingly, however, living kidney donations from non-blood-related partners also show high success rates - even with quite different histocompatibility antigens.

According to our model, it is easy to spread information through the transplant network, so there will be more approach to spread information, and get more living kidney source.

4. Conclusion

We use the *mutual compatibility matrix* to estimate the matching level of the donator and recipient and compute it. We use this matching level .in all models computing in this paper.

We build the recipient maximize weight model. and find the potential bottlenecks for efficient organ matching, We modify the recipient maximize weight model ,and gain the max matching level of the donator and recipient model. We use the Instantaneous Solve Algorithm and Instantaneous Solve II Algorithm to solve two model.

taking into account the medical and psychological dynamics of the situation., and Putting the max number, quality and benefit of exchanges as the objective functions, we build the *multi objective programming* model and describe the procedure to maximize the number and quality of exchanges. We use More Instantaneous Solve Algorithm . to solve this model.

We discuss the ethical dimensions of your recommended exchange procedure and your recommended patient strategy, and suggest ways to develop and recruit more altruistic donors.

The results obtained for all these models are very encouraging.. There is some efficient to handle the potential bottlenecks for efficient organ matching and improve the utilization of the kidney exchanges, and offer new way to solve the problem of bottlenecks for efficient organ matching,.

These models' advantage as follow:

1. These models are very practical.
2. These models can be used very convenient.

These models' disadvantage as follow:

1. These models are very complex.

Of course, if you use the database of the recipient to solve this problem ,you can feel very easy .

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6. Appendix

6.1 Appendix 1

Table 1: Transplants by donor and recipient

Year	donor	recipient
1995	11,081	29,036
1996	11,398	32,284
1997	11,701	35,551
1998	12,451	38,741
1999	12,760	41,234
2000	13,612	44,636
2001	14,265	47,672
2002	14,779	50,447
2003	15,137	53,840
2004	16,004	57,910

6.2 Appendix 2

Table 2: Kidney waiting List Patient Characteristics at End of Year, 1995 to 2004

Current Age	Year									
	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004
Total	25,509	28,775	31,679	34,528	36,981	39,989	42,269	44,337	45,644	45,676
<1 Year	-	1	2	2	1	1	-	-	3	3
1-5 Years	68	59	65	61	48	65	68	64	59	77
6-10 Years	86	95	105	91	92	105	114	114	86	96
11-17 Years	316	361	352	391	363	391	429	467	412	444
18-34 Years	5,401	5,709	5,980	6,083	6,199	6,164	6,213	6,310	6,354	6,137
35-49 Years	10,219	11,215	12,062	12,618	13,002	13,587	13,837	14,179	14,385	13,931
50-64 Years	7,809	9,309	10,629	12,292	13,720	15,419	16,660	17,796	18,463	18,701
65+ Years	1,610	2,026	2,484	2,990	3,556	4,257	4,948	5,407	5,882	6,287
Total (%)	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%
<1 Year	-	0.00%	0.00%	0.00%	0.00%	0.00%	-	-	0.00%	0.00%
1-5 Years	0.30%	0.20%	0.20%	0.20%	0.10%	0.20%	0.20%	0.10%	0.10%	0.20%
6-10 Years	0.30%	0.30%	0.30%	0.30%	0.20%	0.30%	0.30%	0.30%	0.20%	0.20%
11-17 Years	1.24%	1.25%	1.11%	1.13%	0.98%	0.98%	1.01%	1.05%	0.90%	0.97%
18-34 Years	21.17%	19.84%	18.88%	17.62%	16.76%	15.41%	14.70%	14.23%	13.92%	13.44%
35-49 Years	40.06%	38.97%	38.08%	36.54%	35.16%	33.98%	32.74%	31.98%	31.52%	30.50%
50-64 Years	30.61%	32.35%	33.55%	35.60%	37.10%	38.56%	39.41%	40.14%	40.45%	40.94%
65+ Years	6.31%	7.04%	7.84%	8.66%	9.62%	10.65%	11.71%	12.20%	12.89%	13.76%

6.3 Appendix 3

Table 3: One Year Unadjusted Graft Survival by Organ and Year of Transplant, 1994 to 2003

		Year of Transplant									
		1994	1995	1996	1997	1998	1999	2000	2001	2002	2003
Number of Transplants	Kidney: Deceased Donor	7,533	7,599	7,595	7,637	7,893	7,914	7,958	8,068	8,287	8,389
	Kidney: Living Donor	3,007	3,386	3,669	3,927	4,419	4,716	5,487	6,035	6,240	6,468
	Pancreas Transplant Alone	38	36	44	64	72	123	118	127	139	117
	Pancreas After Kidney	54	67	112	130	157	221	303	306	376	343
	Kidney-Pancreas: Kidney	746	910	848	841	967	931	907	886	902	865
	Kidney-Pancreas: Pancreas	746	910	848	841	967	931	907	886	902	865
	Liver: Deceased Donor	3,486	3,772	3,857	3,924	4,269	4,338	4,384	4,453	4,692	5,039
	Liver: Living Donor	56	53	62	86	91	251	392	515	361	318
	Intestine	22	44	41	61	66	67	76	110	103	109
	Heart	2,310	2,336	2,311	2,253	2,293	2,136	2,161	2,169	2,108	2,022
	Lung	707	846	788	908	837	863	940	1,034	1,028	1,065
	Heart-Lung	71	69	38	62	46	51	46	27	31	28
	Heart-Lung	71	69	38	62	46	51	46	27	31	28
Survival (%)	Kidney: Deceased Donor	84.40%	86.30%	87.80%	88.70%	89.00%	88.10%	88.00%	89.00%	89.00%	89.00%
	Kidney: Living Donor	92.70%	92.80%	93.80%	94.20%	94.70%	94.60%	94.20%	94.40%	94.90%	95.30%
	Pancreas Transplant Alone	67.80%	63.90%	72.30%	68.50%	79.20%	83.30%	75.30%	78.30%	79.30%	74.40%
	Pancreas After Kidney	70.40%	70.10%	66.70%	74.40%	72.00%	80.30%	74.00%	81.80%	77.30%	77.90%
	Kidney-Pancreas: Kidney	85.80%	89.20%	89.70%	91.90%	91.30%	91.80%	92.80%	92.00%	91.40%	91.90%
	Kidney-Pancreas: Pancreas	80.70%	82.40%	83.50%	84.50%	82.80%	83.30%	84.20%	85.50%	85.50%	86.00%
	Liver: Deceased Donor	76.40%	77.70%	76.30%	78.60%	79.90%	79.70%	80.80%	80.50%	82.40%	82.00%
	Liver: Living	64.30%	75.50%	85.50%	84.90%	70.30%	74.10%	77.80%	80.20%	79.80%	84.00%

	Donor										
	Intestine	59.10%	56.80%	61.00%	52.50%	51.50%	49.30%	68.40%	62.50%	69.90%	76.80%
	Heart	83.50%	84.00%	84.90%	84.70%	85.00%	83.30%	85.20%	85.30%	86.20%	87.40%
	Lung	74.40%	75.80%	70.50%	75.80%	75.40%	75.50%	77.10%	77.30%	80.50%	82.40%
	Heart-Lung	66.20%	78.20%	65.80%	59.70%	54.30%	56.90%	63.00%	74.10%	61.30%	50.00%
Standard Error(Pct. Pts.)	Kidney: Deceased Donor	0.40%	0.40%	0.40%	0.40%	0.40%	0.40%	0.40%	0.30%	0.30%	0.30%
	Kidney: Living Donor	0.50%	0.40%	0.40%	0.40%	0.30%	0.30%	0.30%	0.30%	0.30%	0.30%
	Pancreas Transplant Alone	7.70%	8.00%	6.80%	5.80%	4.80%	3.40%	4.00%	3.70%	3.50%	4.10%
	Pancreas After Kidney	6.20%	5.60%	4.50%	3.90%	3.60%	2.70%	2.50%	2.20%	2.20%	2.30%
	Kidney-Pancreas: Kidney	1.30%	1.00%	1.00%	0.90%	0.90%	0.90%	0.90%	0.90%	0.90%	0.90%
	Kidney-Pancreas: Pancreas	1.40%	1.30%	1.30%	1.20%	1.20%	1.20%	1.20%	1.20%	1.20%	1.20%
	Liver: Deceased Donor	0.70%	0.70%	0.70%	0.70%	0.60%	0.60%	0.60%	0.60%	0.60%	0.50%
	Liver: Living Donor	6.40%	5.90%	4.50%	3.90%	4.80%	2.80%	2.10%	1.80%	2.10%	2.10%
	Intestine	10.50%	7.50%	7.60%	6.40%	6.20%	6.10%	5.30%	4.60%	4.50%	4.10%
	Heart	0.80%	0.80%	0.70%	0.80%	0.70%	0.80%	0.80%	0.80%	0.80%	0.70%
	Lung	1.60%	1.50%	1.60%	1.40%	1.50%	1.50%	1.40%	1.30%	1.20%	1.20%
	Heart-Lung	5.60%	5.00%	7.70%	6.20%	7.30%	6.90%	7.10%	8.40%	8.70%	9.40%